Next Steps in Finite Lattice-to-$\overline{\text{MS}}$ Matching

Updated October 12, 2026: Paired Fit-Window Diagnostic

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Purpose and present status

The October 6 morning audit established that the radial counterterm basis, continuum pole residues, coefficient-input interface, and prerequisite diagnostics are ready. The morning audit itself did not evaluate a finite Brillouin-zone difference. A subsequent, deliberately narrow calculation evaluated one scalar $O(4)$ quartic tadpole contribution to the mass matching. That example verifies the first integral and its normalization; it is not a complete mass coefficient, a complete radial matching, or full $SU(2)$ matching. This update specifies a proposed nonlinear lattice $SU(2)$ action whose quadratic expansion agrees with the recorded link Hessian. The existing determinant and fluctuation modes still use quadratic operators, but an exact nonlinear vertex module is now connected to the solver and can generate the proposed rules. The derivation below completes Steps 1 and 2 for the explicitly selected scalar-only radial one-loop scope, including its momentum-space potential vertices and finite lattice–$\overline{\text{MS}}$ differences. On October 8, a fixed full-gauged nonexceptional routing and operator-projector deck was generated and independently audited. The selected vertex tensors, routing checks, local projectors, and tree-level off-shell Higgs-slope projector pass their stated tests. This completes the routing and projector-definition portion of full-gauged Step 1. A first Step 2 calculation now evaluates the one-loop Higgs two-point function at the symmetric background in $\xi = 1$, together with scalar-only four- and six-point projections at the fixed nonexceptional routings. A scalar-matter gauge-polarization Ward check also passes within its integration error. A first gauge-linked four-radial-Higgs insertion sum has now been evaluated as a pilot at the same fixed routes. The October 11 reproduction samples six common IR-regulator values through $\lambda_{\text{IR}}/\mu_R = 1/256$ with eight replicates of 2^{12} points per route and regulator. The quadratic fit uses the first three values and holds the three lower-regulator values out as checks. The insertion sum passes its determinant, link-vertex, symmetry, and UV-tail checks, but the quoted uncertainties are QMC-only and do not estimate extrapolation-model error. These remain sector checks, not the full set of gauged multi-point amplitudes or the full gauge/ghost Ward system. An October 12 paired fit-window diagnostic evaluates three-, four-, and five-regulator quadratic fits on the saved lower-statistics, seed-nested QMC prefix. The window shifts are not resolved from zero at the paired QMC precision, while the same-sign lower-regulator residuals persist at about 1.4 paired standard errors. This does not estimate extrapolation-model uncertainty. The coefficient template therefore remains unpopulated and its full-matching flag must remain false.

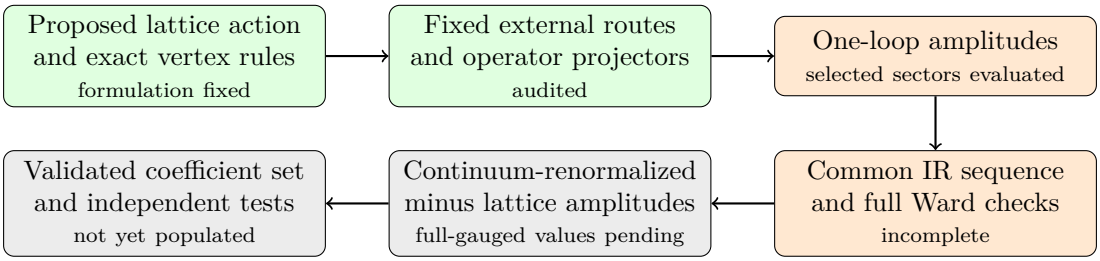


Figure 1: Status map for finite matching. Green boxes denote fixed and audited setup, orange boxes denote work with partial sector results, and gray boxes denote downstream results that must remain open until the full-gauged amplitudes and regulator checks are complete.

1 Lattice action and momentum kernel

The solver uses a forward nearest-neighbor difference for the radial kinetic term. In four dimensions its discretized form is

$$S_{\mathrm{kin}}^{\mathrm{lat}} = \frac{a^4}{2} \sum_x \sum_{\nu=1}^{4} \left(\frac{\rho_{x+a\hat{\nu}} - \rho_x}{a} \right)^2 .$$

The same difference acts componentwise for the four real components of the Higgs multiplet. Its free momentum-space kernel is

$$\widehat{k}^{\,2} = \frac{4}{a^2} \sum_{\nu=1}^{4} \sin^2\!\left(\frac{a k_\nu}{2} \right), \qquad k_\nu \in [-\pi/a, \pi/a].$$

Thus the massive scalar propagator is

$$G_{\mathrm{lat}}(k) = \frac{1}{\widehat{k}^{\,2} + m^2}, \qquad m^2 = -\texttt{mu_sq} > 0$$

at the false vacuum. In the solver defaults, $a = 0.8$ and $m^2 = 0.4$. The counterterm stage default matching scale is

$$\mu = \sqrt{0.4} = 0.6324555320\ldots,$$

not 10.4. The external momentum p used to project a two-point function is a separate variable from the renormalization scale μ.

2 Proposed complete nonlinear lattice action

The existing fluctuation and determinant operators implement the scalar forward-difference action and the quadratic gauge–Goldstone–ghost Hessian; they do not evaluate the full nonlinear action in loop amplitudes. The solver now also exposes exact nonlinear vertex generators and a table-generation mode for the following proposed Wilson-action completion. This fixes the matching convention without claiming that the full-gauged loop calculation has been incorporated into the production determinant.

Let $T^a = \sigma^a/2$, g_0 be the bare lattice coupling, and place a fundamental Higgs doublet H_x on each site. Define the link and plaquette variables by

$$U_{x\mu} = \exp[-iag_0 A^a_{x\mu} T^a], \tag{1}$$

$$U_{x,\mu\nu} = U_{x\mu} U_{x+a\hat{\mu},\nu} U^\dagger_{x+a\hat{\nu},\mu} U^\dagger_{x\nu}. \tag{2}$$

The logarithmic gauge field is understood on the perturbative branch around $U_{x\mu} = 1$, which is the branch used to derive the matching Feynman rules; explicitly, $A^a_{x\mu} = 2i\,\mathrm{Tr}[T^a \log U_{x\mu}]/(ag_0)$. Use the Wilson plaquette action

$$S_g = \frac{4}{g_0^2} \sum_x \sum_{\mu<\nu} \left(1 - \frac{1}{2}\,\mathrm{ReTr}\,U_{x,\mu\nu}\right).$$

Its continuum normalization is $\frac{1}{4}\int d^4x\, F^a_{\mu\nu}F^a_{\mu\nu}$ for the generator convention above; this is the Wilson plaquette formulation [1].

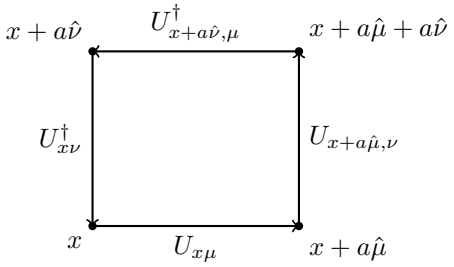


Figure 2: One oriented Wilson plaquette. The arrows follow the ordered product $U_{x,\mu\nu}$; daggered factors traverse links opposite to their positive coordinate direction. Its weight in the action is $\frac{4}{g_0^2}(1 - \frac{1}{2}\,\mathrm{ReTr}\,U_{x,\mu\nu})$.

Normalize the doublet by $\rho_x = \sqrt{2H_x^\dagger H_x}$ and use the covariant nearest-neighbor scalar action

$$S_{H,\mathrm{link}} = a^4 \sum_{x,\mu} \frac{(U_{x\mu}H_{x+a\hat{\mu}} - H_x)^\dagger (U_{x\mu}H_{x+a\hat{\mu}} - H_x)}{a^2}.$$

At $U = 1$, write $H_x = (\phi_x^2 + i\phi_x^1, \phi_x^4 - i\phi_x^3)^T/\sqrt{2}$; this gives $\frac{1}{2}a^4 \sum_{x,\mu,A}[(\phi^A_{x+a\hat{\mu}} - \phi^A_x)/a]^2$, the canonical scalar normalization used in the solver. The full tree potential is

$$V_0(\rho) = -\frac{1}{2}\mu_{\mathrm{sq}}\rho^2 + \frac{1}{4}\lambda\rho^4 + \frac{1}{6}\gamma\rho^6 + \frac{1}{8}\delta\rho^8, \qquad m_0^2 = -\mu_{\mathrm{sq}}.$$

Here μ_{sq} denotes the solver input named `mu_sq`; it is negative at the metastable origin, so $m_0^2 > 0$. Include the dimension-10 and dimension-12 operators as perturbative matching counterterms, not as leading tree interactions:

$$V_{\mathrm{ct}}(\rho) = \frac{1}{2}\delta m^2\rho^2 + \frac{1}{4}\delta\lambda\rho^4 + \frac{1}{6}\delta\gamma\rho^6 + \frac{1}{8}\delta\delta\rho^8 + \frac{1}{10}\delta c_{10}\rho^{10} + \frac{1}{12}\delta c_{12}\rho^{12}.$$

The gauge-covariant kinetic counterterm is δZ_H times $S_{H,\mathrm{link}}$.

For perturbation theory around a radial background $\bar{\rho}_x$ with vanishing background gauge field, choose background R_ξ gauge

$$F^a_x = (DA^a)_x - \xi q \bar{\rho}_x \chi^a_x, \qquad S_{\mathrm{gf}} = \frac{a^4}{2\xi}\sum_{x,a}(F^a_x)^2, \qquad q = \frac{g_0}{2},$$

where D is the backward divergence already defined in the solver documentation. Define the Faddeev–Popov operator by the exact variation of this gauge condition under the lattice transformations

$$H_x \mapsto G_x H_x, \qquad U_{x\mu} \mapsto G_x U_{x\mu} G^\dagger_{x+a\hat{\mu}}, \qquad G_x \in SU(2),$$

and include $S_{\rm gh} = a^4 \sum_{x,y,a,b} \bar{c}^a_x M^{ab}_{xy} c^b_y$ with $M^{ab}_{xy} = \delta F^a_x / \delta \omega^b_y$ at zero gauge transformation, using $G_x = \exp[-ig_0 \omega^a_x T^a]$. This definition fixes the nonlinear ghost vertices once the link-field coordinate and gauge-fixing function are expanded consistently. The gauge-fixing and ghost terms here define the perturbative background-field calculation; they are not part of the gauge-invariant bare action.

The complete proposed perturbative action is therefore

$$S = S_g + S_{H,\rm link} + a^4 \sum_x [V_0(\rho_x) + V_{\rm ct}(\rho_x)] + S_{\rm gf} + S_{\rm gh}.$$

Keep the coupling in the link variables as g_0 until the bare-to-renormalized coupling matching is specified. The benchmark value $g(\mu) = 0.65$ is not by itself a determination of $g_0(a)$.

2.1 Quadratic check against the implemented Hessian

Expanding $S_{H,\rm link}$ about $H_x = (0, \bar{\rho}_x)^T/\sqrt{2}$ and $H_x = (\chi_x, \bar{\rho}_x + h_x)^T/\sqrt{2}$ in a single real generator plane gives, for each Goldstone direction,

$$S^{(2)}_{H,\rm link} = \frac{a^4}{2} \sum_{x,\mu} \left[\frac{(\chi_{x+a\hat{\mu}} - \chi_x)^2}{a^2} + q^2 \bar{\rho}_x \bar{\rho}_{x+a\hat{\mu}} A^2_{x\mu} \right] \tag{3}$$

$$+ a^4 \sum_{x,\mu} \frac{q}{a} A_{x\mu} (\bar{\rho}_{x+a\hat{\mu}} \chi_x - \bar{\rho}_x \chi_{x+a\hat{\mu}}). \tag{4}$$

This agrees with the link-Hessian expression retained in the validation report. At $\bar{\rho} = 0$, the Higgs and three Goldstones have mass m_0, and the gauge fields and ghosts are massless. With $\widehat{k}_\mu = 2\sin(ak_\mu/2)/a$, the free kernels are

$$G_{AB}(k) = \frac{\delta_{AB}}{\widehat{k}^{\,2} + m_0^2}, \tag{5}$$

$$K_{\mu\nu}(k) = \widehat{k}^{\,2} \delta_{\mu\nu} - (1 - \xi^{-1}) \widehat{k}_\mu \widehat{k}_\nu, \tag{6}$$

$$D_{\mu\nu}(k) = \frac{1}{\widehat{k}^{\,2}} \left[\delta_{\mu\nu} - (1 - \xi) \frac{\widehat{k}_\mu \widehat{k}_\nu}{\widehat{k}^{\,2}} \right], \qquad G_{\rm gh}(k) = \frac{1}{\widehat{k}^{\,2}}. \tag{7}$$

For the link-midpoint Fourier convention and $\xi = 1$, $D_{\mu\nu} = \delta_{\mu\nu}/\widehat{k}^{\,2}$. The zero-momentum massless gauge and ghost modes require a declared infrared and gauge-volume prescription. The following subsections give exact momentum-space generators for the nonlinear gauge, scalar-link, and ghost vertices from S. They are implemented in `su2_lattice_vertices.py` and imported by the solver; the existing Hessian code alone does not supply them. They generate the displayed one-loop vertex tables, but are not yet consumed by the matching integrator or the physical determinant calculation.

2.2 Step 1 derivation: exact vertex generators

The following expressions define the lattice vertices without replacing the link action by a continuum derivative expansion. Write $H_+ = H_{x+a\hat{\mu}}$ and $A = A^a_{x\mu} T^a$. For one oriented link, the scalar action is exactly

$$S_{H,x\mu} = a^2 \left[H^\dagger_+ H_+ + H^\dagger_x H_x - H^\dagger_+ e^{+iag_0 A} H_x - H^\dagger_x e^{-iag_0 A} H_+ \right].$$

Thus every scalar–scalar–r-gauge-field vertex is generated by the rth derivative of the two matrix exponentials at $A = 0$. For example, the terms linear and quadratic in the link field are

$$S^{(A)}_{H,x\mu} = ia^3 g_0 \left(H^\dagger_x A H_+ - H^\dagger_+ A H_x \right), \tag{8}$$

$$S^{(A^2)}_{H,x\mu} = \frac{a^4 g_0^2}{2} \left(H^\dagger_+ A^2 H_x + H^\dagger_x A^2 H_+ \right). \tag{9}$$

Fourier transformation must retain the endpoint phases; the link field is assigned to the link midpoint, yielding the factors $\widehat{k}_\mu = 2\sin(ak_\mu/2)/a$ in the quadratic rules.

Likewise, all pure-gauge vertices are generated by differentiating

$$\frac{4}{g_0^2} \sum_{x,\mu<\nu} \left[1 - \frac{1}{2} \mathrm{ReTr} \prod_{\ell \in \partial p_{x,\mu\nu}} e^{-iag_0 \epsilon_\ell A^a_\ell T^a} \right],$$

where $\epsilon_\ell = +1$ for a positively oriented link and -1 for its inverse. The product is path ordered as in the plaquette definition. This prescription fixes the three- and four-gauge vertices (and higher vertices), including their lattice momentum phases, as in perturbative lattice vertex generation [2]. The ghost vertices are the functional derivatives of $M = \delta F/\delta\omega$ from the specified link transformation and field coordinate; they cannot be obtained from the quadratic ghost operator alone.

For the scalar potential, let $X = \phi^A \phi^A$ and

$$f(X) = -\frac{1}{2}\mu_{\mathrm{sq}} X + \frac{1}{4}\lambda X^2 + \frac{1}{6}\gamma X^3 + \frac{1}{8}\delta X^4.$$

All scalar potential vertices follow from $V_{A_1 \cdots A_n} = \partial_{A_1} \cdots \partial_{A_n} f(X)$. An explicit compact rule is

$$V_{A_1 \cdots A_n} = \sum_{j=0}^{\lfloor n/2 \rfloor} 2^{n-j} f^{(n-j)}(X) \sum_{P_j} \left(\prod_{\{r,s\} \in P_j} \delta_{A_r A_s} \right) \left(\prod_{t \text{ unpaired in } P_j} \phi_{A_t} \right),$$

where P_j runs over all choices of j disjoint pairs from the n indices. This includes the radial and transverse curvatures

$$U_L = m_0^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \qquad U_T = \frac{1}{\rho}\frac{dV_0}{d\rho} = m_0^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6,$$

with their smooth limits at $\rho = 0$. These are vertex rules derived from the proposed action, not vertices already implemented or validated in code.

2.3 Reduced momentum-space rules for the nonlinear vertices

For the rules in this subsection use

$$\varphi_x = \int_{\mathrm{BZ}} \frac{d^4p}{(2\pi)^4} e^{ip\cdot x} \varphi(p)$$

for site fields and the same convention with x replaced by the link midpoint $x + a\hat{\mu}/2$ for $A_{x\mu}$. All momenta are incoming, and each vertex below is the reduced action-density vertex after factoring out the periodic momentum-conservation delta. Momentum is conserved modulo a reciprocal lattice vector. This convention fixes the endpoint and midpoint phases and avoids treating a continuum derivative rule as an exact lattice vertex.

For one scalar link, take incoming Fourier labels p' on $H^\dagger$, p on H, and $k_1, \dots, k_r$ on gauge legs, so $p' + p + \sum_i k_i = 0$. Define

$$\theta_\mu = \frac{a}{2}(p' - p)_\mu, \qquad \mathcal{T}^{a_1 \cdots a_r} = \frac{1}{r!} \sum_{\pi \in S_r} T^{a_{\pi(1)}} \cdots T^{a_{\pi(r)}}, \qquad \mathcal{T}^{\varnothing} = \mathbf{1}.$$

The reduced vertex with all gauge legs polarized along the link direction μ is

$$\mathcal{V}^{a_1 \cdots a_r}_{H^\dagger H A^r_\mu} = -\frac{(iag_0)^r}{a^2} \left[e^{i\theta_\mu} + (-1)^r e^{-i\theta_\mu} \right] \mathcal{T}^{a_1 \cdots a_r}.$$

For any other polarization assignment this single-link contribution vanishes. The first rules are

$$\mathcal{V}_{H^\dagger H} = \widehat{p}^2, \qquad \mathcal{V}^a_{H^\dagger H A^a_\mu} = \frac{2g_0}{a} \sin\theta_\mu \, T^a, \qquad \mathcal{V}^{ab}_{H^\dagger H A^a_\mu A^b_\mu} = \frac{g_0^2}{2} \cos\theta_\mu \, \delta^{ab} \mathbf{1}.$$

The last expression is the second functional derivative, so it includes both assignments of the two labeled external gauge legs. The formula at general r follows by differentiating the exact link exponentials; it retains their lattice phases and gives every scalar–scalar–r-gauge vertex, rather than only the continuum-limit $H^\dagger H A$ rule.

For the pure-gauge vertices, label the four ordered edges of a $\mu\nu$ plaquette by $j = 1, \dots, 4$. Their axes, orientation signs, and midpoint offsets from the plaquette origin are

$$(\alpha_j) = (\mu, \nu, \mu, \nu), \quad (\epsilon_j) = (+1, +1, -1, -1), \quad (d_j) = \left(\frac{a\hat{\mu}}{2}, a\hat{\mu} + \frac{a\hat{\nu}}{2}, a\hat{\nu} + \frac{a\hat{\mu}}{2}, \frac{a\hat{\nu}}{2} \right).$$

For n labeled incoming gauge legs (a_i, μ_i, k_i), assign each leg i to an edge σ_i with $\alpha_{\sigma_i} = \mu_i$. For each edge let $I_j = \{i : \sigma_i = j\}$, $m_j = |I_j|$, and set

$$B_j(I_j) = \frac{(-iag_0\epsilon_j)^{m_j}}{m_j!} \sum_{\pi \in S_{m_j}} T^{a_{\pi(1)}} \cdots T^{a_{\pi(m_j)}}, \qquad B_j(\varnothing) = \mathbf{1}.$$

Then the reduced n-gauge vertex is

$$\mathcal{V}^{a_1 \cdots a_n}_{\mu_1 \cdots \mu_n}(k_1, \dots, k_n) = -\frac{2}{a^4 g_0^2} \sum_{\mu < \nu} \sum_{\substack{\sigma_i \in \{1,2,3,4\} \\ \alpha_{\sigma_i} = \mu_i}} \left(\prod_{i=1}^n e^{ik_i \cdot d_{\sigma_i}} \right) \mathrm{ReTr}[B_1(I_1) B_2(I_2) B_3(I_3) B_4(I_4)] \, .$$

The prefactor uses the action density $S_g = a^4 \sum_x (S_g/a^4)$; the displayed trace is the coefficient of the specified labeled legs. This ordered-edge formula generates the quadratic, cubic, quartic, and higher Wilson vertices, including their color factors and exact lattice phases. For checking color algebra one may use

$$\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}, \quad \mathrm{tr}(T^a T^b T^c) = \frac{i}{4}\epsilon^{abc}, \quad \mathrm{tr}(T^a T^b T^c T^d) = \frac{1}{8}(\delta^{ab}\delta^{cd} - \delta^{ac}\delta^{bd} + \delta^{ad}\delta^{bc}).$$

The quadratic vertex summed over plaquettes gives the transverse lattice kernel; adding the stated gauge-fixing term gives $K_{\mu\nu}$ above.

The ghost rules are most compactly and unambiguously specified by the exact Jacobian of the logarithmic link coordinate. Write $t^a = -iT^a$, $X = ag_0 A^a t^a = \log U$, and let $C = \mathrm{ad}_X$; in real adjoint components, $(CY)^a = ag_0 \epsilon^{acb} A^c Y^b$. Under $U_{x\mu} \mapsto G_x U_{x\mu} G^\dagger_{x+a\hat{\mu}}$, $G_x = \exp(g_0 \omega^a_x t^a)$, differentiation of the matrix exponential gives the exact link-coordinate variation

$$\delta X = \frac{C}{e^C - 1} \left[g_0 \omega_x - e^C g_0 \omega_{x+a\hat{\mu}} \right].$$

Equivalently,

$$\delta A_{x\mu} = \frac{1}{a}\Bigg[(\omega_x - \omega_{x+a\hat{\mu}}) - \frac{ag_0}{2}A_{x\mu} \times (\omega_x + \omega_{x+a\hat{\mu}}) \tag{10}$$

$$+\frac{(ag_0)^2}{12}A_{x\mu} \times (A_{x\mu} \times (\omega_x - \omega_{x+a\hat{\mu}})) - \frac{(ag_0)^4}{720}A^{\times 4}_{x\mu}(\omega_x - \omega_{x+a\hat{\mu}}) + \cdots\Bigg], \tag{11}$$

where $A \times Y$ denotes the adjoint cross product and $A^{\times 4}Y = A \times (A \times (A \times (A \times Y)))$. More generally, the exact function $C/(e^C - 1)$ generates all powers and fixes the link ghost vertices for this logarithmic field coordinate.

With $H = (\phi^2 + i\phi^1, \phi^4 - i\phi^3)^T/\sqrt{2}$, denote $\chi^a = \phi^a$ and $h = \phi^4 - \bar{\rho}$. The infinitesimal Higgs transformation is

$$\delta\chi^a = q[-(\bar{\rho} + h)\delta^{ab} + \epsilon^{abc}\chi^c]\omega^b, \qquad \delta h = q\chi^b\omega^b, \qquad q = g_0/2.$$

Thus the full Faddeev–Popov vertex generator is

$$M^{ab}_{xy} = \sum_\mu D_{x\mu}\frac{\delta A^a_{x\mu}}{\delta\omega^b_y} - \xi q\bar{\rho}_x\frac{\delta\chi^a_x}{\delta\omega^b_y}, \qquad S_{\rm gh} = a^4\sum_{x,y}\bar{c}^a_x M^{ab}_{xy}c^b_y.$$

For constant $\bar{\rho}$ and vanishing fluctuation fields this gives

$$M^{ab}_0(k) = \delta^{ab}(\widehat{k}^{\,2} + \xi q^2\bar{\rho}^2),$$

and the scalar ghost vertices are

$$\bar{c}^a c^b h: \quad +\xi q^2\bar{\rho}\,\delta^{ab}, \qquad \bar{c}^a c^b\chi^c: \quad -\xi q^2\bar{\rho}\,\epsilon^{abc}.$$

For clarity, with incoming antighost momentum p, ghost momentum k, and gauge momentum $-p-k$, the one-gauge ghost vertex is

$$\mathcal{V}_{\bar{c}^a c^b A^c_\mu} = -ig_0\epsilon^{abc}\widehat{p}_\mu\cos\!\left(\frac{ak_\mu}{2}\right), \qquad \widehat{p}_\mu = \frac{2}{a}\sin\!\left(\frac{ap_\mu}{2}\right).$$

The higher $\bar{c}cA^r$ vertices follow by inserting the higher terms of the exact logarithmic variation into $D\delta A/\delta\omega$; in particular the two-gauge ghost rule comes from the C^2 term together with the link endpoint phases. This is also a practical consistency check on the link-field coordinate: using a linearized continuum transformation instead would change these vertices.

2.4 Step 1 local validation

The solver exposes the exact generators through its imported vertex module. Its `su2_vertices` table-generation mode produces the complete color and polarization tensors for scalar-link $r = 0, 1, 2$, Wilson-gauge $n = 2, 3, 4$, gauge-ghost $r = 0, 1, 2$, and the local $\bar{c}ch$ and $\bar{c}c\chi$ vertices at requested momentum sets. The saved sample tables cover every index component at their listed routings; the arbitrary-momentum functions remain available for other routings. The standalone audit listed below checks every saved scalar, Wilson, and gauge-ghost table entry. It obtains scalar entries by finite differencing the exact link exponentials, Wilson entries by a separate ordered-word expansion and checks leg permutation/quadratic-kernel identities, and gauge-ghost entries by finite differences of the exact logarithmic link coordinate under finite transformations.

It also compares the complete nonzero-background sparse Faddeev–Popov matrix on a periodic 2^4 lattice against finite gauge transformations with nonzero links, Goldstones, radial fluctuations, and a spatially varying gauge-fixing background. The maximum absolute discrepancies at $a = 0.8$, $g_0 = 0.65$, and $\xi = 1$ were:

Check	Max. abs. discrepancy
Momentum conservation for all generated tables	0
Scalar-link Taylor coefficients (orders 1–2)	8.020×10^{-9}
Complete scalar-link momentum/color table	6.204×10^{-9}
Wilson-plaquette Taylor coefficients (orders 2–4)	1.719×10^{-9}
Complete Wilson momentum/color tensor vs. independent expansion	0
Wilson leg-permutation symmetry	4.442×10^{-16}
Wilson quadratic kernel vs. transverse Hessian	7.216×10^{-16}
Log-link gauge variation	5.954×10^{-11}
Real-Higgs transformation	1.813×10^{-10}
Scalar-link gauge-invariance Ward check	1.776×10^{-15}
Plaquette gauge-invariance Ward check	2.220×10^{-16}
Complete gauge-ghost momentum/color table vs. finite transformations	2.398×10^{-6}
Local scalar-ghost tensors	1.735×10^{-18}
Nonzero-background FP matrix (maximum absolute)	3.234×10^{-10}
Nonzero-background FP matrix (relative)	1.679×10^{-11}
Free Faddeev–Popov kernel	7.105×10^{-15}

The complete arrays and their momentum metadata are saved in `su2_nonlinear_vertex_tables_20261007_conserving.npz` and `su2_nonlinear_vertex_tables_20261007_conserving.json`. The checks validate every color/polarization entry for those selected kinematic routings and validate the nonzero-background FP Jacobian against an independent finite transformation. They are numerical implementation checks, not perturbative uncertainty estimates; they do not sample every possible external momentum. The nonlinear vertex APIs are implemented. The fixed routing/projector and common-IR prescription for the next full-gauged calculation are now recorded below; the full-gauged matching integrator, loop amplitudes, regulator extrapolation, and finite matching coefficients remain outstanding.

2.5 October 8 fixed full-gauged routing and projector deck

The reproducible deck is generated by `su2_full_gauged_projection_setup.py`; the arrays and routing metadata are retained in `su2_full_gauged_projection_tables_20261008.npz` and `su2_full_gauged_projection_tables_20261008.json`. All momenta are incoming, site fields use $e^{+ip\cdot x}$, and gauge fields are located at link midpoints. The selected setup is $a = 0.8$, $g_0 = 0.65$, $\xi = 1$, $\mu_R^2 = 0.4$, and the illustrative local-vertex background is $\bar{\rho} = 0.37$.

For off-shell two-point functions use the common direction $n = (1, 2, 3, 4)/\sqrt{30}$ and $p = \sqrt{s}\, n$ at $s = p^2 \in \{0.2, 0.4, 0.6\}$, with the corresponding legs $(p, -p)$. At the subtraction point $p^2 = \mu_R^2 = 0.4$, the lattice coordinate is $\widehat{p}^2 = 0.3966567139724254$. The asymmetric scalar–scalar–gauge triangle uses momenta $p_{H^\dagger} = \sqrt{0.4}(1, 0, 0, 0)$, $p_H = \sqrt{0.4}(0.2, 0.4, 0.8, 0)$, and $p_A = -p_{H^\dagger} - p_H$; its squared norms are 0.4, 0.336, and 0.896, so no leg is exceptional. The symmetric three-gauge/ghost triangle and four-leg tetrahedron are fixed by the stored arrays and each has per-leg continuum $p_i^2 = 0.4$. For local four-, six-, eight-, ten-, and twelve-scalar contacts, fixed-shape zero-sum routings are stored at three average squared-momentum scales $\bar{s} = 0.2, 0.4, 0.6$; every proper subset sum is nonzero. The metadata is the authoritative full-precision momentum record.

For the Higgs two-point function, fit the three samples to $c_0 + c_1 x + c_2 x^2$, using $x = p^2$ in the continuum and $x = \widehat{p}^2$ on the lattice. The off-shell slope projector at the subtraction point is the row $(0, 1, 2x_\mu)$ times the inverse design matrix; the quadratic term absorbs leading fixed-shape momentum

contamination. It returns unit slope for both continuum and lattice tree kernels to 1.11×10^{-15}. This validates the projector algebra only: applying it to the one-loop $\Sigma^{(2)}$ is Step 2 and has not been done.

For a gauge two-point tensor, let $K = p$ in the continuum and $K_\mu = 2\sin(ap_\mu/2)/a$ on the lattice. The orthogonal projectors contract against $T_{\mu\nu} = K^2\delta_{\mu\nu} - K_\mu K_\nu$ and $L_{\mu\nu} = K_\mu K_\nu$:

$$P_T^{\mu\nu} = \frac{T^{\mu\nu}}{3(K^2)^2}, \qquad P_L^{\mu\nu} = \frac{K_\mu K_\nu}{(K^2)^2}.$$

For each local $2n$-scalar contact, contract the complete tensor first, then fit its three fixed-shape samples to $d_0 + d_1\bar{s} + d_2\bar{s}^2$ and retain the intercept d_0. Separate continuum and lattice design matrices use $\bar{s} = p^2$ and $\bar{s} = \widehat{p}^{\,2}$ respectively. This intercept is the local contact projector under the stated truncation; it is not a claim that momentum-dependent operators have been calculated.

Use background Feynman gauge on both regularizations and add the same auxiliary $\lambda_{\rm IR}^2$ to every massless gauge and ghost denominator:

$$k^2 \mapsto k^2 + \lambda_{\rm IR}^2, \qquad \widehat{k}^{\,2} \mapsto \widehat{k}^{\,2} + \lambda_{\rm IR}^2, \qquad \lambda_{\rm IR}/\mu_R \in \{1/8, 1/16, 1/32\}.$$

The massive scalar denominators are unshifted. These are fixed common regulator settings for Step 2. They are used below in the Higgs two-point gauge diagrams, where the continuum-minus-lattice projected slope is extrapolated to $\lambda_{\rm IR} = 0$. This tests that selected sector only; the regulator dependence and any regulator-induced Ward-identity breaking of the omitted pure-gauge and ghost loops have not been tested.

The independent auditor `su2_full_gauged_projection_audit.py` reports PASS; its machine-readable record is `su2_full_gauged_projection_audit_20261008.json`. Key maximum discrepancies are:

Audit item	Maximum discrepancy
Momentum conservation / smallest proper-subset norm	4.65×10^{-16} / 7.14×10^{-2}
Scalar-link tensors vs. independent finite-action differences	2.90×10^{-9}
Asymmetric scalar–scalar–gauge Ward identity	3.82×10^{-17}
Wilson tensors vs. independent ordered-word expansion	1.33×10^{-15}
Wilson two-point transversality / gauge projectors	6.11×10^{-16} / 2.19×10^{-15}
Ghost tensors vs. exact finite transformations	2.72×10^{-6}
Nonzero-background Faddeev–Popov matrix (absolute / relative)	3.23×10^{-10} / 1.68×10^{-11}
Higgs tree-slope / local-contact projector identities	1.11×10^{-15} / 8.88×10^{-16}

These tests cover the selected complete color/polarization tensors and the listed identities; they do not amount to an exhaustive momentum sweep, a one-loop Ward-identity test, or full-gauged matching.

3 Step 2 derivation: one-loop projections

At one loop, the zero-derivative background-field contribution is generated by the bosonic and ghost Hessians. For a constant radial background ρ, zero background gauge field, and $\xi = 1$, set $M_A^2 = q^2\rho^2$ and $\widehat{k}^{\,2} = 4a^{-2}\sum_\mu \sin^2(ak_\mu/2)$. Up to a ρ-independent constant, the lattice one-loop potential is

$$V_{1,\rm lat}(\rho) = \frac{1}{2}\int_{\rm BZ} \frac{d^4k}{(2\pi)^4}\left[\log(\widehat{k}^{\,2} + U_L) + 3\log(\widehat{k}^{\,2} + U_T + M_A^2) + 6\log(\widehat{k}^{\,2} + M_A^2)\right].$$

The first term is the radial mode, the second the three Goldstones, and the last is the net gauge-plus-ghost contribution for three colors in Feynman gauge. The continuum comparison replaces the

Brillouin-zone integral by the $d = 4 - 2\epsilon$ continuum integral and retains the d-dependent vector/ghost multiplicity through $\overline{\text{MS}}$ subtraction; explicitly, before subtraction it is

$$V_{1,\text{cont}}(\rho) = \frac{1}{2}\left(\frac{e^{\gamma_E}\mu_R^2}{4\pi}\right)^{\epsilon} \int \frac{d^d k}{(2\pi)^d} \left[\log(k^2 + U_L) + 3\log(k^2 + U_T + M_A^2) + 3(d-2)\log(k^2 + M_A^2)\right].$$

Setting the vector/ghost multiplicity to its four-dimensional value before subtracting can lose a finite term. The determinant construction follows the background-field effective-action framework [3]; the familiar nonanalytic gauge contribution is also present in the Coleman–Weinberg effective potential [4].

For the two-point function, define the kinetic projection from the momentum dependence of the one-loop self-energy, using $\widehat{p}^2$ on the lattice side and p^2 in the continuum:

$$\delta Z_H(\mu) = \frac{\partial}{\partial p^2}\Sigma^{(2)}_{\overline{\text{MS}},\text{ren}}(p)\Big|_{p^2=\mu^2} - \frac{\partial}{\partial \widehat{p}^2}\Sigma^{(2)}_{\text{lat}}(p)\Big|_{\widehat{p}^2=\mu^2}.$$

Mass and higher potential coefficients are projections of $\Gamma^{(2n)}_{\overline{\text{MS}},\text{ren}} - \Gamma^{(2n)}_{\text{lat}}$ onto the local $O(4)$-invariant operators, using identical external momenta and an operator projector that separates potential terms from momentum-dependent operators. The matching sign agrees with the scalar tadpole example: continuum-renormalized amplitude minus lattice amplitude.

> *The constant-background determinant is a useful check, but its Taylor expansion at $\rho = 0$ is not by itself a valid full-gauge-theory potential projector: massless gauge modes generate nonanalytic terms proportional to $\rho^4 \log \rho^2$. The common infrared prescription, nonexceptional Euclidean routings, and local projector definitions are now fixed above. Regulated one-loop numbers have been obtained only for the Higgs two-point slope and scalar-only four- and six-point cross-checks. The gauge/Goldstone/ ghost contributions to the multi-point integrals, and the full massless-sector infrared cancellation, remain unevaluated. No full-gauged finite coefficient is inferred from the determinant formula or the partial checks. The nonanalyticity is the standard one-loop gauge contribution to the effective potential [4].*

Step 1–2 status (updated October 8). The selected scalar-only radial potential calculation below remains the completed radial result. For the full-gauged program, Step 1's fixed routing, projectors, selected complete vertex tensors, and tree-level checks are recorded in Section 2.5. Step 2 is now partially evaluated: the one-loop Higgs $\Sigma^{(2)}$ slope has been calculated with the common three-point λ_{IR} sequence and extrapolated to zero; scalar-only nonexceptional four- and six-point amplitudes have been projected as cross-checks; and the scalar-matter part of the gauge-polarization Ward identity has been tested. Full-gauged multi-point amplitudes, the pure-gauge and ghost Ward identities, a systematic regulator/model-error study, and the finite full-gauged matching coefficients remain open. The coefficient JSON remains null.

3.1 Evaluated one-loop sectors at the fixed routing

The calculation is implemented in `su2_one_loop_step2_partial_20261008.py`; its machine-readable output is `su2_one_loop_step2_partial_20261008_corrected.json`. The lattice and continuum calculations use $a = 0.8$, $g_0 = 0.65$, $m_0^2 = \mu_R^2 = 0.4$, $\lambda = -2$, and $\xi = 1$. The run uses scrambled Sobol Brillouin-zone integration for the lattice, dimensional regularization with $\overline{\text{MS}}$ subtraction for the continuum, and the fixed off-shell projector at $p^2 = 0.2, 0.4, 0.6$.

At $\bar{\rho} = 0$, the one-loop Higgs two-point diagrams are the scalar quartic tadpole and gauge exchange plus seagull; the Faddeev–Popov operator has no Higgs coupling at this background in the selected gauge. Adding the same λ_{IR}^2 to the internal gauge denominator on both sides gives:

$\lambda_{\mathrm{IR}}/\mu_R$	$\delta Z_H^{\overline{\mathrm{MS}}-\mathrm{lat}}$	QMC standard error
1/8	0.0328817051	3.41×10^{-7}
1/16	0.0329036915	3.45×10^{-7}
1/32	0.0329097345	3.55×10^{-7}

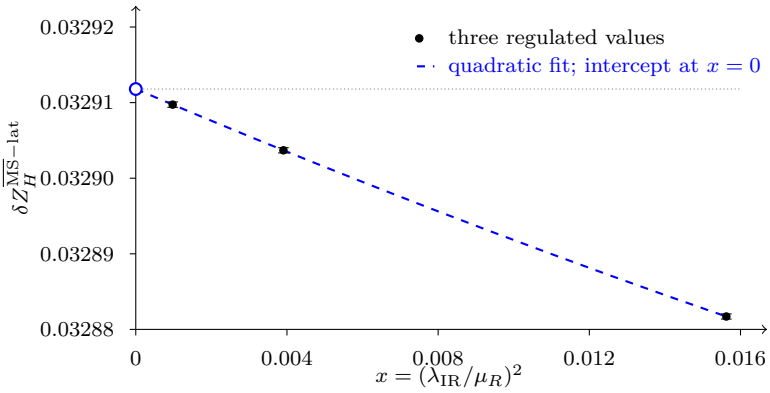


Figure 3: Higgs-slope finite difference versus the squared common IR regulator. Vertical bars show the listed QMC standard errors; the open point and dotted line mark the reported zero-regulator intercept. The quadratic is fixed by exactly these three points, so the graph does not supply an independent extrapolation-model check.

A quadratic fit in $(\lambda_{\mathrm{IR}}/\mu_R)^2$ gives the intercept $\delta Z_H(0) = 0.0329117974$ with a QMC-only standard error 3.60×10^{-7}. Because three regulator points determine the quadratic fit exactly, this quoted error excludes extrapolation-model uncertainty; there is no independent fourth-point fit check. The monotone approach across the listed sequence is evidence of regulator stability for this two-point combination, not a proof of cancellation in the uncomputed gauge/ghost sectors.

The stored local contact projectors also give scalar-only fixed-routing cross-checks:

Amplitude	Projected continuum-minus-lattice coefficient	QMC error
Four scalar legs	−1.4510393	5.16×10^{-5}
Six scalar legs	−2.8468383	1.76×10^{-4}

These finite-momentum scalar checks include no gauge-linked diagrams. Their $O(4)$ scalar trace already includes all four real scalar components, including the three Goldstone directions. Missing from the full-gauged Higgs amplitudes are the scalar-link $hA\chi$ mixing and $hhAA$ seagull contributions. In particular the six-point intercept is sensitive to omitted higher momentum dependence in the three-scale quadratic fit; neither entry should be substituted for a full-gauged coefficient or treated as an exact zero-momentum radial result. The scalar loops have massive internal lines, so they have no λ_{IR} dependence.

At this symmetric point the chosen gauge-fixing function is $F^a = \partial \cdot A^a$. Its Faddeev–Popov operator depends on the gauge field but not on the Higgs background, so derivatives of the one-loop ghost

determinant with respect to Higgs-only sources vanish. Thus no direct ghost loop is missing from these Higgs-only amplitudes at $\bar{\rho} = 0$; ghost loops remain necessary in gauge-field amplitudes, including the gauge two-point function.

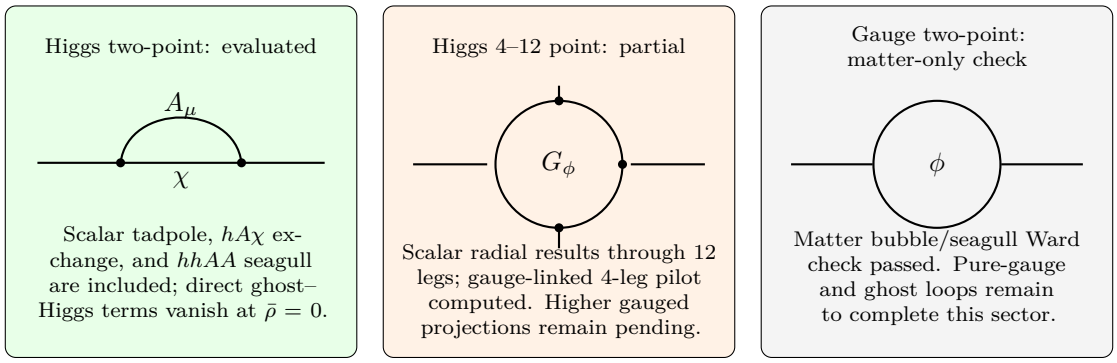


Figure 4: One-loop topology and completion status at the symmetric background. The sketches indicate diagram families rather than full momentum routings or combinatorial weights. The four-leg gauge-linked panel denotes a pilot projection with limited QMC/extrapolation checks; the higher scalar-only projections still lack their gauge-linked derivatives.

For the remaining Higgs-only amplitudes, organize the one-loop bosonic functional using the coupled gauge–scalar Hessian in a general radial source $h(x) = \phi^4(x)$:

$$\Gamma_B^{(1)}[h] = \frac{1}{2}\,\mathrm{Tr}\log\mathcal{K}_B[h], \qquad \mathcal{K}_B[h] = \begin{pmatrix} K_A^{(0)} + \Pi_{AA}[h^2] & V_{A\phi}[h] \\ V_{\phi A}[h] & K_\phi^{(0)} + U''(he_4) \end{pmatrix}.$$

The block dimensions are twelve gauge and four real scalar components. $V_{A\phi}[h]$ is generated by the scalar-link $hA\chi$ vertex and $\Pi_{AA}[h^2]$ by its $hhAA$ seagull. The scalar block contains all three Goldstone channels and the radial channel, with potential curvatures

$$U_T''(h) = m_0^2 + \lambda h^2 + \gamma h^4 + \delta h^6, \qquad U_L''(h) = m_0^2 + 3\lambda h^2 + 5\gamma h^4 + 7\delta h^6.$$

The full Higgs $2n$-point integrands are the $2n$th source derivatives of $\Gamma_B^{(1)}[h]$ at $h = 0$, evaluated at the fixed external momenta. This may be evaluated as the finite insertion expansion

$$\Gamma_B^{(1)}[h] - \Gamma_B^{(1)}[0] = \frac{1}{2}\sum_{j\geq 1}\frac{(-1)^{j+1}}{j}\,\mathrm{Tr}\Big[(G_0\,\Delta\mathcal{K}_B[h])^j\Big], \quad G_0 = \mathcal{K}_B[0]^{-1}, \quad \Delta\mathcal{K}_B[h] = \mathcal{K}_B[h] - \mathcal{K}_B[0].$$

This specifies the bosonic diagram sum. The gauge-linked four-radial-Higgs derivative has now been integrated at the fixed routes as a pilot described below; the corresponding six-, eight-, ten-, and twelve-point derivatives remain to be added to their scalar-only projections.

At the fixed $p^2 = 0.4$ gauge momentum, the scalar-matter contribution to the gauge polarization has a continuum normalized transverse residual 1.15×10^{-15}. On the lattice the normalized residual is 4.77×10^{-4}, while the four contracted components are within 1.06 QMC standard errors of zero. This is a matter-loop Ward check only; it omits pure-gauge and ghost loops and therefore does not test the complete one-loop Slavnov–Taylor identities. Since this diagram has only massive scalar internal propagators, it does not require the massless λ_{IR} regulator.

3.2 Pilot gauge-linked four-Higgs projection

The first missing multi-point sector has been evaluated for four external radial Higgs fields at the fixed local-contact routes. The insertion sum contains the $hhAA$–$hhAA$ term (PP), the two-mixing plus

seagull terms (PVV), the two-mixing plus scalar-potential-Hessian terms (UVV), and the four-mixing term (VVVV), where $V = hA\chi$, $P = hhAA$, and U is the λh^2 Goldstone-Hessian insertion. Direct Higgs–ghost diagrams vanish at the symmetric background in the stated gauge. The continuum integrand is MS-bar subtracted by removing

$$\frac{C}{(k^2+\mu_R^2)^2}, \qquad C = -54q^4 + 36q^2\lambda, \qquad q = \frac{g_0}{2},$$

whose renormalized integral is zero at μ_R. For the benchmark parameters $C = -8.2074586$. The lattice and continuum amplitudes use the same three fixed external shapes and the same common mass regulator on every internal gauge line. The projected continuum-minus-lattice four-point contributions are:

$\lambda_{\mathrm{IR}}/\mu_R$	$\Delta\Gamma^{(4)}_{\mathrm{gauge}}$	QMC standard error
1/8	0.2542605	3.34×10^{-4}
1/16	0.2541718	5.43×10^{-4}
1/32	0.2540645	6.81×10^{-4}
1/64	0.2539542	7.53×10^{-4}
1/128	0.2538960	7.83×10^{-4}
1/256	0.2538767	7.92×10^{-4}

A quadratic fit in $(\lambda_{\mathrm{IR}}/\mu_R)^2$ to the first three values gives a zero-regulator intercept 0.2540212 ± 0.0007361 (QMC standard error only). The three lower-regulator points are out-of-fit checks: observed minus predicted is $-7.82 \times 10^{-5} \pm 5.45 \times 10^{-5}$ at 1/64, $-1.28 \times 10^{-4} \pm 9.03 \times 10^{-5}$ at 1/128, and $-1.45 \times 10^{-4} \pm 1.03 \times 10^{-4}$ at 1/256. Each is about 1.4 QMC standard errors below the quadratic prediction. The residual errors include cross-regulator covariance from paired Sobol scrambles, but these same-sign residuals do not bound extrapolation-model error. The preserved four-point-fit, lower-statistics comparison gives an intercept 0.2539852 ± 0.0016464; because it uses a seed-nested sample, it is a fit-window diagnostic rather than an independent reproduction. The October 12 paired audit extends that lower-statistics, seed-nested sequence to three-, four-, and five-regulator quadratic fits. Their intercepts are 0.2541409 ± 0.0015671, 0.2539852 ± 0.0016464, and 0.2538930 ± 0.0016921, respectively (replicate QMC errors only). Since these fits use the same six randomized replicates, the paired shifts are more informative than subtracting the marginal errors: the four-point minus three-point shift is -0.0001557 ± 0.0001113, and the five-point minus three-point shift is -0.0002479 ± 0.0001753. Both are about 1.4 paired QMC standard errors below zero, so the window shifts are not statistically resolved by this lower-statistics sample. The three-point fit's held-out residuals at 1/64, 1/128, and 1/256 are respectively -0.0002190 ± 0.0001565, -0.0003620 ± 0.0002543, and -0.0004110 ± 0.0002872 (paired QMC errors), again about 1.4 standard errors below their quadratic predictions. This same-sign pattern is a fit-window warning, not a bound on extrapolation-model error. The audit reproduces the earlier four-point-fit intercept exactly, but it does not provide paired errors for the October 11 eight-replicate, 2^{12}-point central values. With the normalization $\delta\lambda = \Delta\Gamma^{(4)}/3!$, the gauge-sector local-contact diagnostic from the primary fit is $+0.04234 \pm 0.00012$ (QMC standard error only); it is not entered into the full-matching coefficient file.

The insertion sum reproduces the constant-source gauge/Goldstone determinant to below 7×10^{-13}, its nonzero-momentum $hA\chi$ and $hhAA$ rules agree with the exact link generator to below 6×10^{-17}, and the large-momentum integrand reproduces C/k^4 to relative error 8.1×10^{-7}. The projected integrand is Bose symmetric to numerical precision. The October 11 reproduction uses eight independently scrambled Sobol replicates with 2^{12} points per route and regulator. All quoted standard errors are QMC-only replicate errors; they do not include extrapolation-model uncertainty, perturbative

truncation, or other systematics. This remains a pilot and does not provide a complete gauge/ghost Slavnov–Taylor check. The high-statistics output and evaluator are `su2_four_point_gauge_sector_20261011_ircheck.json` and `su2_four_point_gauge_sector_20261008.py`. The October 8 baseline is `su2_four_point_gauge_sector_20261008_ircheck.json`. A lower-statistics, same-seed-prefix run with a four-point fit is preserved in `su2_four_point_gauge_sector_20261011_extended_ircheck.json`; it is a fit-window comparison, not an independent reproduction.

4 First evaluated diagram: scalar quartic tadpole

Take the scalar interaction to be

$$V_4(\Phi) = \frac{\lambda}{4}(\Phi^A \Phi^A)^2, \qquad A = 1, \ldots, 4.$$

For an $O(N)$ multiplet in this normalization, the zero-momentum one-loop tadpole self-energy is $(N+2)\lambda I$. With $N = 4$, it is $\Sigma_{\text{tad}} = 6\lambda I$. This is only the quartic scalar tadpole; it omits other scalar interactions, gauge and ghost loops, and any higher one-particle-irreducible diagrams or operator projections.

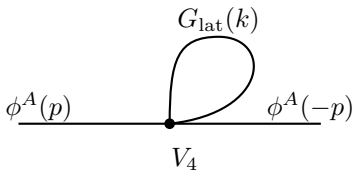


Figure 5: Scalar quartic tadpole contributing to the Higgs two-point function. The loop contraction gives $(N+2)\lambda I$ for the interaction normalization used here; it is only one contribution to the mass matching.

Writing $q_\nu = ak_\nu$ and $x = a^2m^2$, the lattice integral is

$$I_{\text{lat}}(a,m) = \int_{[-\pi/a,\pi/a]^4} \frac{d^4k}{(2\pi)^4} \frac{1}{\widehat{k}^{\,2} + m^2} \tag{12}$$

$$= \frac{1}{a^2} \int_{[-\pi,\pi]^4} \frac{d^4q}{(2\pi)^4} \frac{1}{4\sum_\nu \sin^2(q_\nu/2) + x} \tag{13}$$

$$= \frac{1}{a^2} J(x), \quad J(x) = \int_0^\infty dt\, e^{-xt} \left[e^{-2t} I_0(2t)\right]^4, \tag{14}$$

where I_0 is the modified Bessel function. The final representation is a one-dimensional, absolutely convergent integral for $x > 0$ and was used for the numerical evaluation.

In the same Euclidean convention, the renormalized continuum tadpole after $\overline{\text{MS}}$ subtraction is

$$I_{\overline{\text{MS}}}(m,\mu) = \frac{m^2}{16\pi^2}\left[\ln\left(\frac{m^2}{\mu^2}\right) - 1\right].$$

Define the additive mass matching convention by equating the zero-momentum two-point functions,

$$m^2_{\text{lat}} + \Sigma_{\text{lat}}(0) = m^2_{\overline{\text{MS}}} + \Sigma^{\text{ren}}_{\overline{\text{MS}}}(0), \qquad m^2_{\text{lat}} = m^2_{\overline{\text{MS}}} + \delta m^2_{\text{tad}}.$$

For this diagram and convention only,

$$\delta m^2_{\text{tad}} = 6\lambda \left(I_{\overline{\text{MS}}} - I_{\text{lat}}\right).$$

At $a = 0.8$, $m^2 = \mu^2 = 0.4$, and $\lambda = -2$, numerical quadrature gives

Quantity	Value in solver units
$J(a^2m^2)$	0.1452018039938607
$I_{\rm lat}$	0.2268778187404072
$I_{\overline{\rm MS}}$	−0.002533029591058444
$\delta m^2_{\rm tad}$	2.752930179977589

The adaptive quadrature estimate for the absolute error in J was 2.45×10^{-14}, corresponding to about 3.82×10^{-14} in $I_{\rm lat}$ and 4.6×10^{-13} in this tadpole difference. These numerical integration estimates do not include perturbative truncation or model/systematic uncertainties. The large additive shift is a lattice cutoff-dependent mass contribution, not a universal finite constant.

5 What this example does not determine

The result above must not be entered as the solver value of `delta_m2`: it is one diagram contribution only. In particular, the quartic tadpole has no external-momentum dependence and cannot determine the kinetic coefficient `delta_ZH`. Nor does the calculation determine the $\rho^4, \rho^6, \rho^8, \rho^{10}, \rho^{12}$ coefficients, gauge-field normalization, gauge-coupling matching, or a decay-rate prefactor. Keep `includes_full_su2_lattice_to_msbar_matching=false` until the complete matching scope represented by that flag has actually been computed.

6 Completed Steps 1–2: scalar-only radial matching

To make the requested first calculation finite and reproducible, this section selects the scalar-only radial scope: retain the $O(4)$ tree potential through ρ^8, compute its one-loop scalar determinant, and project the local potential basis through ρ^{12}. Gauge, link, and ghost loops are excluded from these numbers. The calculation therefore completes Steps 1 and 2 for this selected scope only; it is neither full $SU(2)$ matching nor a set of production-solver coefficients.

6.1 Step 1: scalar momentum-space vertex deck

Use the Fourier convention $\phi^A_x = \int_{\rm BZ} d^4p/(2\pi)^4\, e^{ip\cdot x}\phi^A(p)$. All momenta are incoming and are conserved modulo a reciprocal lattice vector. The free scalar propagator is

$$G_{AB}(p) = \frac{\delta_{AB}}{\widehat{p}^{\,2} + m_0^2}, \qquad m_0^2 = -\mu_{\rm sq} > 0.$$

For r disjoint index pairs, define

$$\mathcal{P}_r(A_1, \ldots, A_{2r}) = \sum_{\text{pairings } P} \prod_{\{i,j\}\in P} \delta_{A_iA_j}.$$

At vanishing background the nonzero tree potential vertices are momentum-independent and are

$$V^{(2)}_{A_1A_2} = m_0^2 \delta_{A_1A_2}, \tag{15}$$

$$V^{(4)}_{A_1\cdots A_4} = 2\lambda\, \mathcal{P}_2, \tag{16}$$

$$V^{(6)}_{A_1\cdots A_6} = 8\gamma\, \mathcal{P}_3, \tag{17}$$

$$V^{(8)}_{A_1\cdots A_8} = 48\delta\, \mathcal{P}_4. \tag{18}$$

Here each expression is multiplied by the momentum-conservation delta (or its finite-lattice Kronecker counterpart). In general, for a term $c_r(\phi^A\phi^A)^r/(2r)$, its $2r$-point tensor at the origin is $c_r 2^r r!/(2r)$ times the sum over all pairings. Thus the matching counterterm vertices have the same tensor structure with coefficients δm^2, $2\delta\lambda$, $8\delta\gamma$, $48\delta\delta$, $384\delta c_{10}$, and $3840\delta c_{12}$ for respectively 2, 4, 6, 8, 10, and 12 scalar legs. Since these counterterms are one-loop order, they are not inserted in the one-loop propagators or loop vertices.

6.2 Step 2: zero-momentum projector and finite coefficients

All scalar propagators have the positive mass m_0 at the expansion point $\rho = 0$. There is therefore no scalar infrared singularity, and zero external momenta give a valid local-potential projector in this scalar-only calculation. Put $z = \rho^2$, $D(k) = \widehat{k}^{\,2}$, and

$$P_L(z) = 3\lambda z + 5\gamma z^2 + 7\delta z^3, \qquad P_T(z) = \lambda z + \gamma z^2 + \delta z^3.$$

The one-loop lattice potential, up to a z-independent constant, is

$$V_{1,\mathrm{lat}}^{\mathrm{scalar}}(z) = \frac{1}{2}\int_{\mathrm{BZ}} \frac{d^4k}{(2\pi)^4}\left[\log(D + m_0^2 + P_L(z)) + 3\log(D + m_0^2 + P_T(z))\right].$$

The continuum comparison is the same radial-plus-three-transverse scalar determinant in dimensional regularization with $\overline{\mathrm{MS}}$ subtraction at μ_R. Define

$$I_n^{\mathrm{lat}} = \int_{\mathrm{BZ}} \frac{d^4k}{(2\pi)^4}\frac{1}{(\widehat{k}^{\,2} + m_0^2)^n} = \frac{a^{2n-4}}{(n-1)!}\int_0^\infty dt\, t^{n-1}e^{-a^2m_0^2t}\left[e^{-2t}I_0(2t)\right]^4,$$

where I_0 is the modified Bessel function. The continuum-renormalized moments needed through $n = 6$ are

$$I_1^{\overline{\mathrm{MS}}} = \frac{m_0^2}{16\pi^2}\left[\log\left(\frac{m_0^2}{\mu_R^2}\right) - 1\right], \quad I_2^{\overline{\mathrm{MS}}} = -\frac{1}{16\pi^2}\log\left(\frac{m_0^2}{\mu_R^2}\right), \quad I_n^{\overline{\mathrm{MS}}} = \frac{\Gamma(n-2)}{16\pi^2\Gamma(n)}(m_0^2)^{2-n} \quad (n \geq 3).$$

Let $[z^n]Q$ denote the coefficient of z^n in a polynomial Q. Expanding the two logarithms gives the matching coefficient of z^n in $V_{1,\overline{\mathrm{MS}}}^{\mathrm{ren}} - V_{1,\mathrm{lat}}$:

$$\Delta v_n = \frac{1}{2}\sum_{j=1}^{n}\frac{(-1)^{j+1}}{j}\left(I_j^{\overline{\mathrm{MS}}} - I_j^{\mathrm{lat}}\right)[z^n]\left(P_L(z)^j + 3P_T(z)^j\right), \qquad \delta c_{2n} = 2n\,\Delta v_n.$$

For $n = 1, \ldots, 6$, δc_{2n} denotes in order $\delta m^2, \delta\lambda, \delta\gamma, \delta\delta, \delta c_{10}, \delta c_{12}$. This sign convention is continuum-renormalized amplitude minus lattice amplitude, matching the preceding quartic-tadpole check. Equivalently, the zero-momentum $2n$-point projector with all external indices equal to one gives $\delta c_{2n} = \Delta\Gamma^{(2n)}_{1\cdots1}/(2n-1)!$.

For the matching-stage benchmark $a = 0.8$, $m_0^2 = \mu_R^2 = 0.4$, $\lambda = -2$, $\gamma = 0.8$, and $\delta = 0.08$, the moments are:

n	I_n^{lat}	$I_n^{\overline{\mathrm{MS}}}$
1	0.2268778187404072	−0.002533029591058444
2	0.03094303011979438	0
3	0.009469337125601956	0.007915717472057639
4	0.007100564330706442	0.006596431226714698
5	0.008539339433655424	0.008245539033393373
6	0.01264685637877819	0.01236830855009006

Substitution yields the scalar-only one-loop potential coefficients:

Coefficient	Value (solver units)
δm^2	2.75293017998
$\delta\lambda$	−1.45119341289
$\delta\gamma$	−2.85119512150
$\delta\delta$	0.356724233481
δc_{10}	1.07431690553
δc_{12}	6.20491819587

The mass entry reproduces the independently tabulated quartic-tadpole contribution. Adaptive one-dimensional quadrature repeated with relative and absolute tolerances 2×10^{-12} and 2×10^{-13} changed each listed coefficient by less than 1.1×10^{-15}; this is a quadrature stability check, not a perturbative or truncation uncertainty estimate. At this loop order the scalar-only two-point graph at $\rho = 0$ is a momentum-independent quartic tadpole, so its kinetic slope vanishes and $\delta Z_H^{\text{scalar}} = 0$. Gauge contributions to this slope are not included.

These are radial scalar-only matching results at the stated a and scale, not universal constants and not full-$SU(2)$ coefficients. Do not place them in the full-matching coefficient template or set its full-matching flag. The gauge determinant cannot simply be Taylor-expanded at $\rho = 0$ to extend this table: its massless modes require a common IR prescription and an explicit full-gauged operator projector as described above.

7 Corrected next steps

The remaining work is the extension from the completed scalar-only radial calculation above to full gauged matching. The full-gauged Step 1 routing, projector, and selected-vertex audit is complete as described in Section 2.5. The first Step 2 sector checks are reported in Section 3.1; they do not close the full-gauged calculation. The remaining work is:

1. A paired fit-window diagnostic is now available for the four-point pilot; its lower-statistics seed-nested replicate shifts and held-out residuals are reported in Section 3.2. They do not estimate extrapolation-model uncertainty. Continue by extending the insertion expansion to the six-, eight-, ten-, and twelve-scalar projections. The scalar-only $O(4)$ loops already include Goldstones; the missing terms are generated by the scalar-link $hA\chi$ mixing and $hhAA$ seagull. Compute the gauge two-point function with its pure-gauge and ghost loops in addition to the matter contribution, and apply the stored transverse and longitudinal projectors. Direct ghost-Higgs vertices vanish for Higgs-only amplitudes at $\bar{\rho} = 0$ in the stated gauge. Use the same momentum routes and projectors in the lattice and continuum calculations.

2. Apply the common λ_{IR} sequence to every massless gauge and ghost line in the completed amplitudes. Check the full one-loop Ward/Slavnov–Taylor identities, demonstrate regulator cancellation, and use additional regulator points to assess extrapolation-model errors before taking $\lambda_{\text{IR}} \to 0$.

3. Form the finite differences using the explicit sign convention continuum-renormalized amplitude minus lattice amplitude, estimate quadrature and extrapolation errors, and verify the predicted cutoff dependence. Do not require bare mass shifts to be independent of a: power-divergent pieces such as the tadpole mass shift depend on the cutoff. Instead test the matched amplitudes or physical observables under spacing refinement.

4. Only after all coefficients required by the selected scope are available, write a new coefficient JSON using the exact keys `delta_ZH`, `delta_m2`, `delta_lambda`, `delta_gamma`, `delta_delta`, `delta_c10`, and `delta_c12`. Run the basis evaluator on independent smooth backgrounds and retain the full-matching flag as false for any radial-only result.

Full gauge-field and gauge-coupling matching additionally requires suitable nonzero background gauge fields and operator projections. A radial coefficient exercise alone cannot establish a physical decay-rate prefactor; the determinant, collective-coordinate normalization, and controlled continuum and volume limits remain separate requirements.

References

The model-specific expressions and numerical tadpole value in this report are derived from the stated lattice action and the displayed integrals. The publications below provide the underlying lattice and effective-action framework; they do not supply the report's uncomputed finite matching coefficients.

References

[1] K. G. Wilson, "Confinement of quarks," *Physical Review D* **10**, 2445 (1974). doi:10.1103/PhysRevD.10.2445.

[2] M. Lüscher and P. Weisz, "Efficient numerical techniques for perturbative lattice gauge theory computations," *Nuclear Physics B* **266**, 309–356 (1986). doi:10.1016/0550-3213(86)90094-5.

[3] L. F. Abbott, "The background field method beyond one loop," *Nuclear Physics B* **185**, 189–203 (1981). doi:10.1016/0550-3213(81)90371-0.

[4] S. Coleman and E. Weinberg, "Radiative corrections as the origin of spontaneous symmetry breaking," *Physical Review D* **7**, 1888–1910 (1973). doi:10.1103/PhysRevD.7.1888.

Project sources consulted. The implementation and project audits consulted are:

Solver: `Higgs_fluctuation_solver_stabilized.py`, function `make_su2_link_operators`.

Link-Hessian derivation: `SU2_Physical_Prefactor_Progress_Content.tex`.

Measured Hessian checks: `Higgs_SU2_Gauge_Coupled_Hessian_Report_20260930.tex`.

Matching readiness: `SU2_Matching_Coefficient_Progress_Report_20261006.tex`.

Nonlinear-vertex module: `su2_lattice_vertices.py`.

Generated tables: `su2_nonlinear_vertex_tables_20261007_conserving.npz` and metadata `su2_nonlinear_vertex_tables_20261007_conserving.json`.

Independent nonlinear-vertex audit: `su2_nonlinear_vertex_audit.py`.

October 8 fixed full-gauged setup generator and independent auditor: `su2_full_gauged_projection_setup.py` and `su2_full_gauged_projection_audit.py`.

October 8 partial one-loop evaluator and corrected sector results: `su2_one_loop_step2_partial_20261008.py` and `su2_one_loop_step2_partial_20261008_corrected.json`.

Gauge-linked four-Higgs insertion evaluator and October 8 baseline: `su2_four_point_gauge_sector_20261008.py` and `su2_four_point_gauge_sector_20261008_ircheck.json`. October 11 higher-statistics result reported above: `su2_four_point_gauge_sector_20261011_ircheck.json`. The lower-statistics, same-seed-prefix fit-window comparison is `su2_four_point_gauge_sector_20261011_extended_ircheck.json`. The October 12 paired three-/four-/five-window audit is generated by `su2_four_point_fit_window_audit_20261012.py` and recorded in `su2_four_point_fit_window_audit_20261012.json`; it reuses that six-replicate, 2^{11}-point seed prefix and is not an independent high-statistics reproduction.

Fixed arrays, routing metadata, and audit record: `su2_full_gauged_projection_tables_20261008.npz`, `su2_full_gauged_projection_tables_20261008.json`, and `su2_full_gauged_projection_audit_20261008.json`.